# LAB 3 – ARCHITECTURE SELECTION & JUSTIFICATION

## Centralized Audit Trail Compliance Engine

*Problem Statement #50*

### 1. Architecture Selection

We chose Microservices Architecture for the Centralized Audit Trail Compliance Engine. The system is divided into separate services such as Log Ingestion, PII Masking, Audit Storage, Audit Query & Reporting, Export & Verification, and Identity & Access Management.

### 2. Reasons for Choosing Microservices

- The different services can be developed and maintained separately.
- If one service has a problem, the other services can continue to work.

### 3. Security Advantage

The PII Masking Service can mask sensitive information before the audit logs are stored. This helps protect personal information such as credit-card numbers and other PII.

### 4. Performance Benefit

The Audit Query & Reporting Service can be scaled separately when there are many search requests. This can help the system handle large amounts of audit data.

### 5. Components and Interfaces

The component diagram shows the main services and how they communicate using ports, provided interfaces, required interfaces, assembly connectors, and usage dependencies.

### 6. Conclusion

Microservices Architecture is suitable for this system because the main functions can be divided into separate services. This makes the system easier to maintain and allows individual services to handle their own work.

# UML COMPONENT DIAGRAM

